

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	ME-CM
Program Name	Master of Engineering in Construction Engineering and Management
Assessment Year	2024-2025
Program Assessment Coordinator	Ray Lemming, Brent Stephens
Year of Current Assessment Plan	2024-2025
No Assessment This Year per Plan	☐
LO(s) Assessed	2
Presented at program authority meeting date: <i>Please attach minutes</i>	n/a

Methodology

The Master of Engineering in Construction Engineering and Management is a coursework-based graduate degree program that provides students with the knowledge and background that is essential to making decisions at site, company, industrial, and sector levels. Students learn how to plan and schedule projects, estimate and control costs, make economic decisions, administer contracts, organize construction sites, manage construction equipment, analyze productivity, optimize construction activities, plan and manage real estate developments, and address legal problems.

Typically, candidates for this degree have bachelor's degrees in Engineering or Architecture, but individuals in career transition who complete IIT's Certificate Program in Construction Management can also be admitted in furtherance of their education.

The primary learning objectives for this program are for students to:

1. Demonstrate an ability to apply advanced technical and business knowledge and leadership skills to complex construction projects
2. Formulate research questions, conduct literature reviews, and utilize relevant tools to advance knowledge in construction engineering and management

Artifact(s) used to evaluate student learning: Performance on assignments, exams, and deliverables in one of the required courses in the program (typically CAE 570, CAE 571, CAE 574, or CAE 577)

Semester(s) in which artifacts are collected and assessed: End of each academic year

Sample size: Each student in the program who took the selected course

Data to report: Results are reported as statistical distributions (i.e., histograms) and summary statistics for the sample of students in each year on each performance indicator

Results

Data table

A total of 89 students were enrolled in the program in Fall 2024, including 72 who were pursuing the degree as a traditional master's student and 17 who were pursuing it as a co-terminal student. Among these 89 students, 36 took CAE 577 Construction Equipment Management in this academic year (Spring 2025) and are thus used for program assessment in this cycle.

The following artifacts were used to assess performance toward each LO:

- LO #1 (Demonstrate an ability to apply advanced technical and business knowledge and leadership skills to complex construction projects):
 - Mid-term and final exams, which cover technical and business fundamentals and decision-making in construction equipment management
- LO #2 (Formulate research questions, conduct literature reviews, and utilize relevant tools to advance knowledge in construction engineering and management):
 - Final report, which involves formulating research questions, conducting literature reviews, and utilizing relevant tools to advance knowledge in construction engineering and management from a list of predefined topics involving the management of heavy construction equipment for construction projects

Table 1 shows results from the 36 students in the program across LOs #1 and #2. Scores range from 0 to 100 on a fractional basis. Definitions of proficiency levels are as follows: ≥ 0.9 (proficiency), 0.7-0.9 (meets expectations), < 0.7 (does not meet expectations). Distributions of proficiency levels are shown in Figure 1. Artifact samples are archived in Canvas.

Table 1. Results from evaluation rubric applied to LOs #1 and #2 for this academic year

student	LO1	LO1 - DMP	LO2	LO2 - DMP
student 1	0.83	M	0.78	M
student 2	0.96	P	0.94	P
student 3	0.79	M	0.86	M
student 4	0.92	P	0.94	P
student 5	0.97	P	0.77	M
student 6	0.97	P	0.83	M
student 7	0.90	P	0.95	P
student 8	0.89	M	0.95	P
student 9	0.97	P	0.92	P
student 10	0.97	P	0.94	P
student 11	0.90	P	0.95	P
student 12	0.98	P	0.96	P
student 13	0.69	D	0.94	P
student 14	0.97	P	0.93	P
student 15	0.97	P	0.93	P
student 16	0.90	M	0.84	M
student 17	0.80	M	0.94	P
student 18	0.97	P	0.93	P
student 19	0.62	D	0.91	P
student 20	0.47	D	0.85	M

student 21	0.97	P	0.93	P
student 22	0.92	P	0.95	P
student 23	0.97	P	0.90	P
student 24	0.97	P	0.94	P
student 25	0.71	M	0.94	P
student 26	1.00	P	0.94	P
student 27	0.94	P	0.94	P
student 28	0.96	P	0.94	P
student 29	0.97	P	0.90	P
student 30	0.97	P	0.90	M
student 31	0.90	M	0.95	P
student 32	0.97	P	0.94	P
student 33	0.97	P	0.83	M
student 34	0.97	P	0.94	P
student 35	0.97	P	0.89	M
student 36	0.82	M	0.94	P

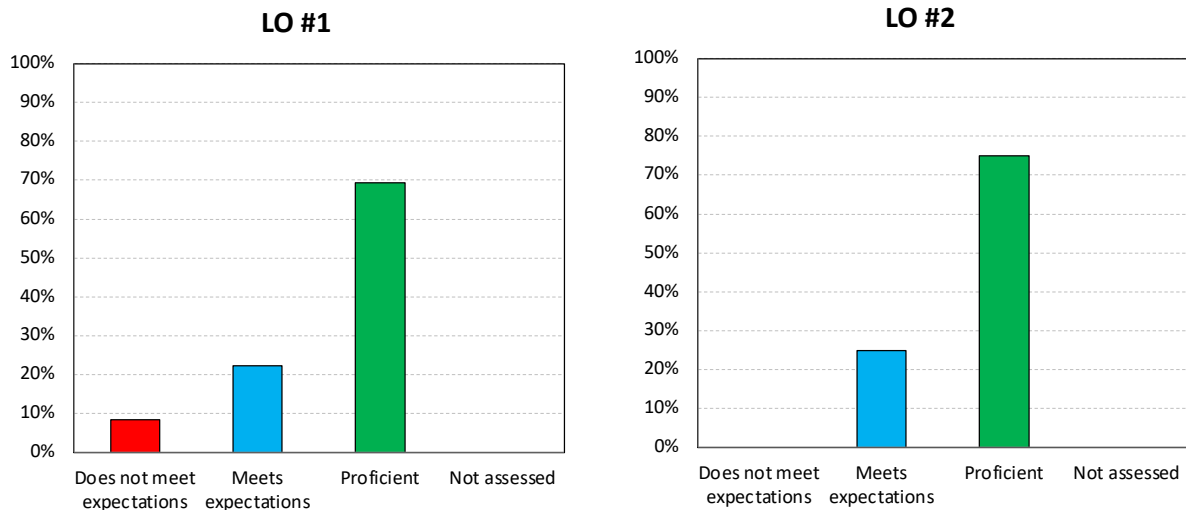


Figure 1. Distribution of proficiency levels for LOs #1, #2, and #3 for this academic year

Description of findings

Key findings from this academic year and cohort include:

- Only 3 out of 36 students sampled did not meet or exceed expectations for LO #1 (8%), suggesting a high degree of proficiency in meeting learning objectives in this cycle.
- All 36 students sampled met or exceeded expectations for LO #2, with a high level of proficiency demonstrated overall (>70% proficient based on final written reports and presentations).
- Combined, a high level of proficiency demonstrated in this course raised no major concerns in this cycle.

Prior CI plan impact

In the previous assessment report, institutional reviewers made several recommendations to improve upon our assessment plan, including converting to an updated format, indicating more clearly which artifacts are used and collected, and to improve upon our CI plan. None of the performance shown above raises major concerns for background preparation for the program or ability to execute in the program.

This cycle's CI plan

1. In this cycle, we revised our learning objectives to 2 simply stated objectives, crafting language that allows for a broader characterization of the primary goals of the program. We also articulated a clearer plan for assessing the program using more than one course if needed in any given year. In this cycle, we selected CAE 577 for assessment since its assignments and mid-term and final exams are well-aligned with preparation for LO #1 while its final project deliverable is well-aligned with preparation for LO #2.